

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
 Manufacturer: Panasonic
 Firmware Version: N/A
 Model(s): PT-FW430U, PT-FW430E, PT-FW430EA, PT-FX400U, PT-FX400E, PT-FX400EA

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.03.0000-b006	2.3.0

Version History

Module Version	Date	Notes
1_1_2_0	5/14/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly. deviceUsername default value is JBMIAPjectorLinkrespectively.
Example: `InterfaceName.devicePassword = 'JBMIAPjectorLinkrespectively'`
- Serial Only:
DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '6', and 'All'.
If DeviceID is set to 'All', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='PT-FW430U')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='PT-FW430U')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='PT-FW430U')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' 'Native' 'V-Fit'	Value 'S4:3' 'Full' 'Wide'	Value 'Normal' 'H-Fit'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1' 'S-Video' 'HDMI'	Value 'Computer 2' 'DVI-I'	Value 'Video' 'Network'
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command LampMode	Value 'Normal'	Value 'Eco'	
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command MenuNavigation	Value 'Menu' 'Down' 'Enter'	Value 'Return' 'Left'	Value 'Up' 'Right'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 63 in steps of 1		
# Volume example InterfaceName.Set('Volume', 63)			

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Auto' 'Native' 'V-Fit'	Value 'S4:3' 'Full' 'Wide'	Value 'Normal' 'H-Fit'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'Computer 1' 'S-Video' 'HDMI'	Value 'Computer 2' 'DVI-I'	Value 'Video' 'Network'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal'	Value 'Eco'	
# LampMode example InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command	Value		

Global Scripter Module Communication Sheet

LampUsage	Hours		
<pre># LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command Power	Value 'On' 'Cooling Down'	Value 'Off'	Value 'Warming Up'
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command Volume	Value 0 to 63 in steps of 1		
<pre># Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>			

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command Input	Value 'Computer 1' 'S-Video' 'HDMI'	Value 'Computer 2' 'DVI-I'	Value 'Video' 'Network'
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Normal' 'Lamp Warning' 'Temperature Error' 'Other Warning'	Value 'Fan Warning' 'Lamp Error' 'Filter Warning' 'Other Error'	Value 'Fan Error' 'Temperature Warning' 'Filter Error' 'Multiple Warnings / Errors'
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command Input	Value 'Computer 1' 'S-Video' 'HDMI'	Value 'Computer 2' 'DVI-I'	Value 'Video' 'Network'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command Power	Value 'On' 'Cooling Down'	Value 'Off'	Value 'Warming Up'
# Power example			

```
InterfaceName.Update('Power')  
Value = InterfaceName.ReadStatus('Power')  
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```


Cable and Adapter Requirements

Captive Screw to Male DB9 RS232 Serial Cable.

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

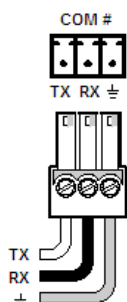
Data Bits: 8

Parity: None

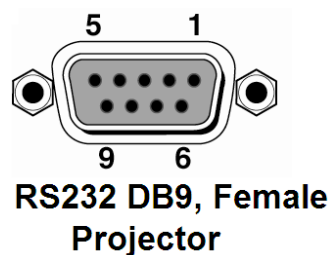
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type:	Ethernet
Default Port:	4352
Logon Credentials Supported:	Yes
Default Password:	JBMIAPjectorLink
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands (Serial)

Aspect Ratio Auto	\x02AD01;VS1:00\x03
Aspect Ratio Full	\x02AD01;VS1:06\x03
Aspect Ratio H-Fit	\x02AD01;VS1:09\x03
Aspect Ratio Native	\x02AD01;VS1:05\x03
Aspect Ratio Normal	\x02AD01;VS1:02\x03
Aspect Ratio S4:3	\x02AD01;VS1:01\x03
Aspect Ratio V-Fit	\x02AD01;VS1:10\x03
Audio Mute Off	\x02AD01;AMT:0\x03
Audio Mute On	\x02AD01;AMT:1\x03
Auto Image None	\x02AD010AS\x03
AV Mute Off	\x02AD01;OSH:0\x03
AV Mute On	\x02AD01;OSH:1\x03
Freeze Off	\x02AD01;OFZ:0\x03
Freeze On	\x02AD01;OFZ:1\x03
Input Computer 1	\x02AD01;IIS:RG1\x03
Input Computer 2	\x02AD01;IIS:RG2\x03
Input DVI-I	\x02AD01;IIS:DVI\x03
Input HDMI	\x02AD01;IIS:HD1\x03
Input Network	\x02AD01;IIS:NWP\x03
Input S-Video	\x02AD01;IIS:SVD\x03
Input Video	\x02AD01;IIS:VID\x03
Lamp Mode Eco	\x02AD01;OLP:0\x03
Lamp Mode Normal	\x02AD01;OLP:1\x03
Menu Navigation Down	\x02AD01;OCD\x03
Menu Navigation Enter	\x02AD01;OEN\x03
Menu Navigation Left	\x02AD01;OCL\x03
Menu Navigation Menu	\x02AD01;OMN\x03
Menu Navigation Return	\x02AD01;OBK\x03
Menu Navigation Right	\x02AD01;OCR\x03
Menu Navigation Up	\x02AD01;OCU\x03
Power Off	\x02AD01;POF\x03
Power On	\x02AD01;PON\x03
Volume 0	\x02AD01;AVL:000\x03
Volume 63	\x02AD01;AVL:063\x03

Appendix B. Update Commands (Serial)

Aspect Ratio	\x02AD01;QS1\x03
AV Mute	\x02AD01;QSH\x03
Freeze	\x02AD01;QFZ\x03
Input	\x02AD01;QIN\x03
Lamp Mode	\x02AD01;QLP\x03
Lamp Usage	\x02AD01;Q\$L\x03
Power	\x02AD01;Q\$S\x03
Volume	\x02AD01;QAV\x03

Appendix C. Set Commands (Ethernet)

AV Mute Off	%1AVMT 30\x0D
AV Mute On	%1AVMT 31\x0D
Input Computer 1	%1INPT 11\x0D
Input Computer 2	%1INPT 12\x0D
Input DVI-I	%1INPT 32\x0D
Input HDMI	%1INPT 31\x0D
Input Network	%1INPT 51\x0D
Input S-Video	%1INPT 22\x0D
Input Video	%1INPT 21\x0D
Power Off	%1POWR 0\x0D
Power On	%1POWR 1\x0D

Appendix D. Update Commands (Ethernet)

AV Mute	%1AVMT ?\x0D
Device Status	%1ERST ?\x0D
Input	%1INPT ?\x0D
Lamp Usage	%1LAMP ?\x0D
Power	%1POWR ?\x0D

